

UTME 2015 BIOLOGY QUESTIONS

- Which of the following has the most primitive respiratory system?
 - insect
 - fish
 - snail
 - mouse
- One adaptation shown by hydrophytes in fresh water habitats is the
 - waxy cuticle on shoot surface
 - poor development of roots and xylem tissues
 - well-developed roots and supporting system
 - leaves reduced to spines
- Which of the following use diffusion as the principal method of gaseous exchange?
 - grasshopper
 - rat spines
 - lizard
 - earthworm
- The theory which supports the view that the large muscles developed by an athlete will be passed on to the offspring was proposed by
 - Mendel
 - Darwin
 - Lamarck
 - Pasteur
- The chromosomes of members of the kingdom Monera are within the
 - nucleoplasm
 - nucleus
 - nucleolus
 - cytoplasm
- The mangrove swamp in Nigeria is restricted to the
 - Sahel savanna
 - Guinea savanna
 - Tropical rainforest
 - Sudan savanna
- The pancreas secretes enzymes for the digestion of
 - fats, proteins and carbohydrates
 - fats, vitamins and cellulose
 - fats, carbohydrates and vitamins
 - proteins, cellulose and minerals
- The causative agent of bird flu is a
 - protozoan
 - virus
 - bacterium
 - fungus
- A water medium is necessary for fertilization in
 - conifers
 - angiosperms
 - ferns
 - fungi
- An example of a sex-linked trait is the
 - colour of the skin in humans
 - ability to roll the tongue
 - possession of facial hair in adult humans
 - ability to grow long hair in females
- In which of the following Nigerian states can montane vegetation be found?
 - Bauchi
 - Plateau
 - Taraba
 - Enugu
- Which of the following is true of cloning?
 - it is welcomed as an ethically and normally sound science
 - it involves the asexual multiplication of the tissues of the original organism
 - the clone is similar to but not exactly like the original organism
 - only one cell of the original organism is needed to imitate the process

13. The process of shedding the exoskeleton of an arthropod is known as

- A. ecdysis
- B. in star formation
- C. metamorphosis
- D. osmosis

14. Which of the following is a major cause of constipation in humans?

- A. lack of roughage
- B. vitamin B
- C. vitamin E
- D. lack of salts

15. In mammals, the organ directly on top of the kidney is the

- A. adrenal gland
- B. prostate gland
- C. pancreas
- D. thyroid gland

16. An accurate identification of a rapist can be carried out by conducting a

- A. RNA analysis
- B. blood group test
- C. behavioural traits test
- D. DNA analysis

17. An example of a fish that aestivates is

- A. croaker
- B. lung fish
- C. shark
- D. cat fish

18. The opening and closing of the stoma are regulated by

- A. respiration
- B. osmosis
- C. diffusion
- D. transpiration

19. Which of the following is common to the mosquito, housefly and blackfly?

- A. they are parasites of man
- B. their immature stages are aquatic
- C. they undergo complete metamorphosis
- D. their adults have two pairs of wings

20. The organs that will be most useful to giant African rats in finding their way in underground habitats are the

- A. nostrils
- B. eyes
- C. vibrissae
- D. tails

21. A crucible of 5gm weighed 10gm after filling with fresh soil. It is then heated in an oven at 1000C for 1 hour. After cooling in a desiccator, the weight was 8gm. The percentage of water in the soil is

- A. 0.8
- B. 0.2
- C. 0.4
- D. 0.6

22. The waste product of plants used in the conversion of hide to leather is

- A. alkaloid
- B. resin
- C. tannin
- D. gum

23. The correct sequence of the movement of urea during formation is

- A. glomerulus - Bowman's capsule - convoluted tubule - Henle's loop - collecting tubule
- B. convoluted tubule - glomerulus - Henle's loop - Bowman's capsule - collecting tubule
- C. glomerulus - Bowman's capsule - convoluted tubule - Henle's loop - collecting tubule

D. convoluted tubule - Bowman's capsule
- Henle's loop - glomerulus - collecting tubule

24. In lizards, the lowering of the gular fold is used to

- A. defend their territory
- B. attract mates
- C. frighten enemies
- D. catch insects

25. The photosynthetic pigments include

- A. chloroplast and cytochromes
- B. melanin and haemoglobin
- C. chlorophyll and carotenoids
- D. carotenoids and haemoglobin

26. The highest level of ecological organization is the

- A. ecosystem
- B. niche
- C. biosphere
- D. population

27. A biotic factor which affects the distribution and abundance of organism in a terrestrial habitat is

- A. pH
- B. competition
- C. temperature
- D. light

28. The eye defect that rises because the cornea is not curved smoothly is

- A. astigmatism
- B. short-sightedness
- C. long-sightedness
- D. presbyopia

29. Which of the following is an example of parasitism?

- A. a squirrel living in an abandoned nest of a bird
- B. mistletoe growing on an orange tree
- C. fungi growing on a dead tree branch

D. cattle egrets taking tasks from the body of cattle

30. The increasing order of the particle size in the following soil types is

- A. cattle sand – clay-gravel
- B. clay - silt sand – gravel
- C. silt - clay - sand - gravel
- D. clay - sand - silt – gravel

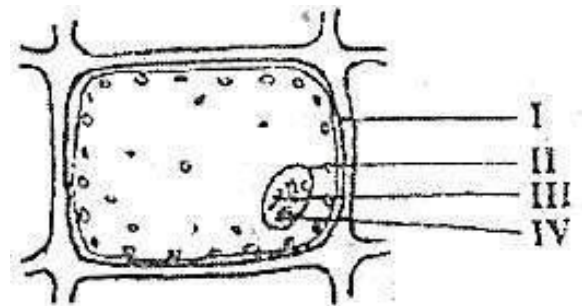
31. Which of following factors can bring about competition population?

- A. emigration
- B. drought
- C. mortality
- D. dispersion

Stunted growth and poor root development are a result of a deficiency in

- A. phosphorus
- B. calcium
- C. sulphur
- D. iron

Use the diagram below to answer question 33 and 34.



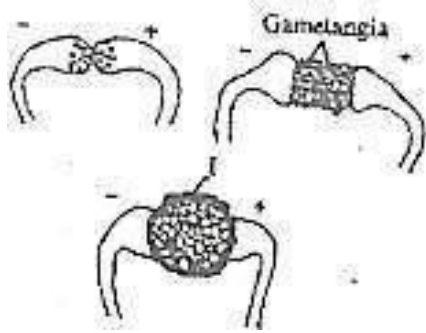
33. The cell organelle solely responsible for respiration is the

- A. nucleus
- B. nucleolus
- C. endoplasmic reticulum
- D. mitochondrion

34. The organelle responsible for heredity is

- A. IV
- B. I
- C. II
- D. III

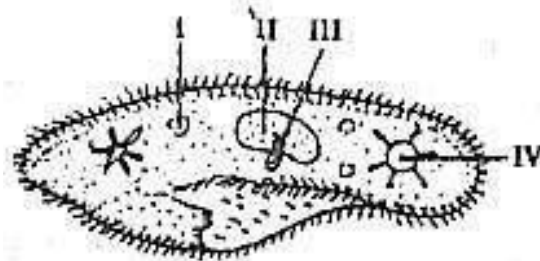
Use the diagram below to answer questions 35 and 36.



35. The process illustrated is
- A. gametogenesis
 - B. sexual reproduction in Rhizopus
 - C. sexual reproduction in Spirogyra
 - D. sporulation

36. The structure labelled I is the
- A. zygospore
 - B. conidiophore
 - C. sporangium
 - D. hypha

Use the diagram below to answer questions 37 and 38.



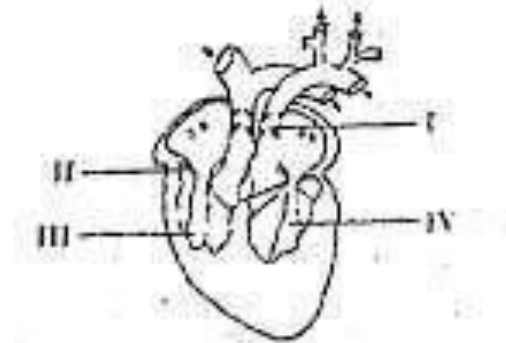
37. The organelle responsible for sexual reproduction is

- A. IV
- B. I
- C. II
- D. III

38. The part labelled IV is responsible for

- A. respiration
- B. ingestion
- C. locomotion
- D. osmoregulation

Use the diagram below to answer questions 39 and 40.



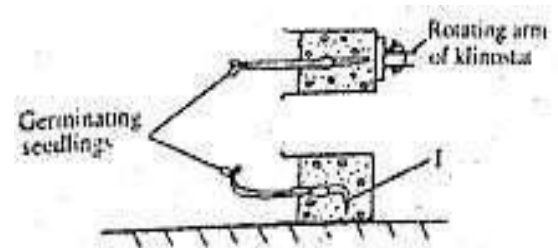
39. The part labelled I is the

- A. pulmonary artery
- B. bicuspid valve
- C. aorta
- D. vena carva

40. Oxygenated blood is pumped to the entire body from the part labelled

- A. IV
- B. I
- C. II
- D. III

Use the diagram below to answer questions 41 and 42.



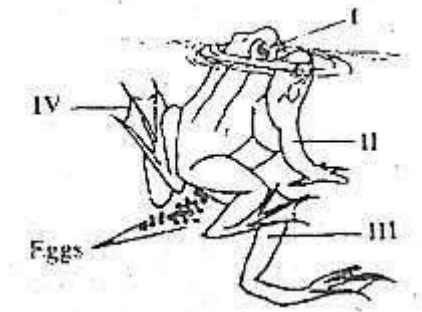
41. The experiment demonstrates

- A. hydrotropism
- B. phototropism
- C. thigmotropism
- D. hydrotropism

42. The part marked I will contain a high concentration of

- A. ethylene
- B. abscisic acid
- C. auxin
- D. ascorbic acid

Use the diagram below to answer questions 43 and 44



43. The breeding posture illustrated in the diagram is known as
- A. reproductive swimming
 - B. amplexus
 - C. mating
 - D. courtship

44. The diagram shows that the organisms are
- A. viviparous
 - B. hermaphrodite
 - C. ovoviviparous
 - D. oviparous

45. Insulin is produced by the endocrine organ labeled
- A. I
 - B. IV
 - C. III
 - D. II

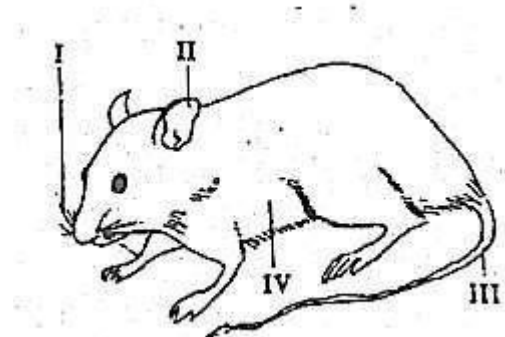
46. Which of the following will be true of dog II which lost its tail in an accident if it mates with dog III?
- A. all its offspring will be born without tails
 - B. 3/4 of its offspring will be born without tails
 - C. none of its offspring will be born without a tail

D. 1/4 of its offspring will be born without tails.

47. If the dogs are offspring of a monohybrid cross and the gene G for grey head is dominant over as allele g, the individual whose genotype is likely to be gg is

- A. I
- B. IV
- C. III
- D. II

Use the diagram below to answer questions 48 and 49.



48. The type of protective adaptation exhibited by the animal is
- A. flash coloration
 - B. countershading colouration
 - C. warning colouration
 - D. disruptive colouration

49. The structure labeled I is
- A. tactile
 - B. radiosensitive
 - C. photosensitive
 - D. chemoreceptive

ANSWER KEYS

1. A

- 2. B
- 3. D
- 4. C
- 5. D
- 6. C
- 7. A
- 8. B
- 9. C
- 10. C
- 11. B
- 12. D
- 13. A
- 14. A
- 15. A
- 16. D
- 17. B
- 18. C
- 19. C
- 20. C
- 21. C
- 22. C
- 23. A
- 24. C
- 25. C
- 26. A
- 27. C
- 28. A
- 29. B
- 30. B
- 31. B
- 32. A
- 33. D
- 34. D
- 35. B
- 36. A
- 37. D
- 38. D
- 39. A
- 40. B
- 41. B
- 42. C
- 43. C
- 44. D
- 45. C
- 46. C

47. C

48. C

49. B

50. A